Literature Status for arXiv:1012.5595: A ZFC Non-Gruenhage Compact Space with (∗)

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Status

This is a `literature_already_answered` packet. It records that Problem 6.2 of Orihuela–Troyanski–Smith, *Strictly convex norms and topology*, arXiv:1012.5595, was answered by Richard J. Smith, *Tree Duplicates, G_δ-diagonals and Gruenhage spaces*, arXiv:1102.0982.

Source Question

Section 6 of arXiv:1012.5595 lists several unresolved problems. The second one, labelled `zfcstarnon-gru`, asks:

> Is there in ZFC an example of a non-Gruenhage compact space with (∗)?

The preceding text explains that the source paper constructs such a compact scattered non-Gruenhage space with (∗) under the continuum hypothesis, but leaves the ZFC version open.

Supporting Answer

The abstract of arXiv:1102.0982 says that the paper gives a ZFC example of a locally compact, scattered, Hausdorff, non-Gruenhage space D having a G_δ-diagonal, and explicitly says that this answers a question posed by Orihuela, Troyanski, and Smith in their work on strictly convex norms.

In the introduction, Smith recalls two topological facts needed to convert this locally compact example into the compact example asked for in Problem 6.2: every space with a G_δ-diagonal has (∗), and if L is locally compact and has (∗), then the one-point compactification $L \cup \{\infty\}$ also has (∗). The main theorem proves that the Λ-duplicate has a G_δ-diagonal but is not Gruenhage. The following corollary then states that the one-point compactification K of that space D is a scattered compact non-Gruenhage space with (∗).

This is exactly the ZFC example requested in Problem 6.2 of arXiv:1012.5595.

Scope

This packet records only the answer to Problem 6.2. It does not address the other problems in Section 6 of arXiv:1012.5595, including the non-scattered strictly convex dual norm problem, the preservation of (∗) under continuous or perfect images, or the Gateaux-norm problem for the Ostaszewski and HMVZ compact O-spaces.

Search Evidence

The lane-13 sparse queue selected arXiv:1012.5595. Cheap run indexes were searched for the arXiv id, title words, `strictly convex norms`, `property (*)`, `non-Gruenhage`, and `G_delta-diagonal`; no duplicate packet for this source was found. Targeted local and web searches for the source problem phrase `non-Gruenhage compact space with (*)` identified arXiv:1102.0982 as the decisive later paper. Its abstract, theorem, and corollary explicitly match the source problem.

References

[1] J. Orihuela, S. Troyanski, and R. J. Smith, *Strictly convex norms and topology*, arXiv:1012.5595.

[2] R. J. Smith, *Tree Duplicates, G_δ-diagonals and Gruenhage spaces*, arXiv:1102.0982.